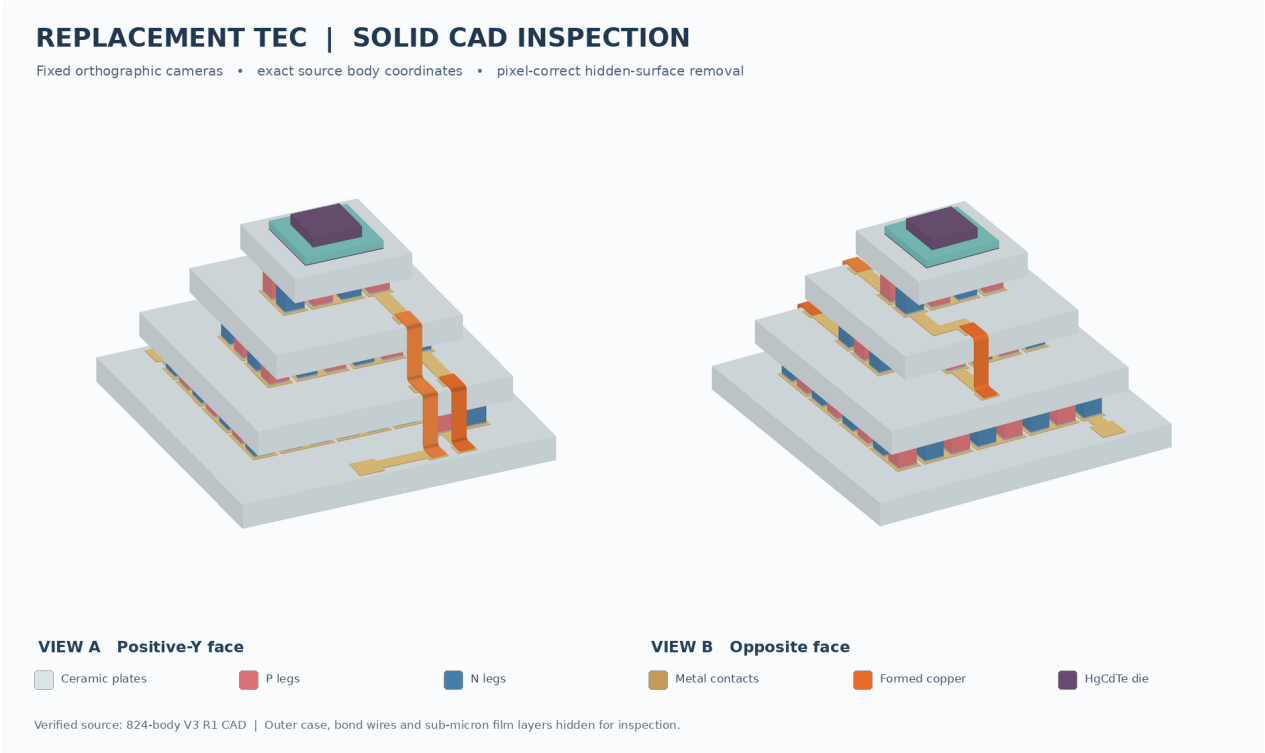
Three-Stage TEC-Cooled

HgCdTe Detector

Current Architecture and Engineering Assessment

Synthetic 824-solid replacement cooler / 88-leg hybrid simulation / 8 November 2026

CONDITIONAL TECHNICAL ASSESSMENT. This is a separate current-architecture document. The historical 38-page Revision 3 report describes an older 216-domain cooler; its scientific methods remain useful, but its internal copper geometry is not the present design. The 43-page interim edition is preserved independently.



Shaded opaque display of the original frozen V3 replacement CAD body primitives and analytical formed-tab bends. The enclosure, external leads and micron-scale layers are hidden for inspection. This remains geometry evidence, not a fully coupled 430-domain PDE or a manufactured device.

Reading contract. All measured/source-backed quantities, selected synthetic construction assumptions, computed numerical outputs and outstanding validation failures are explicitly distinguished.

Executive technical abstract

The modeled article is a plausible but synthetic three-stage thermoelectrically cooled HgCdTe photoconductor in a nine-pin TO-66-style package. Its redesigned physical cooler comprises 824 STEP solids; its derived 430-domain geometry has 2,440 native face regions. It contains 88 series P/N legs (29/11/4 couples) plus short formed interstage tabs and a three-foot stepped return. Independent CAD review supports conditional constructibility, not demonstrated manufacture or reliable electrical/thermal operation.

The current functioning V3.3 solver represents each leg by 32 nonlinear axial transport intervals and resolves four original 3D ceramic P1 FEM fields with 11,791 tetrahedra, plus assumed solder/contact losses, Cu/Au external leads, detector thermal load, hot mounting, effective krypton and gray radiation. For a 0.400 A, 300 K, 2 mW detector, 100 Pa Kr case, it gives

$$\boxed{T_{\mathrm{die}} = 210.625214\ \mathrm{K}}, \quad \boxed{V_{\mathrm{term}} = 1.522462\ \mathrm{V}}, \quad \boxed{P_{\mathrm{in}} = 0.608985\ \mathrm{W}}.$$

The cavity contributes 4.516629 mW into the cold components, with a numerically closed global energy residual of approximately 8.8674×10^{-9} W.

Critical architectural limitation. The formed interstage copper tab and multi-level return geometry has a verified circuit topology, but its nonlocal metal current, contact and thermal-shunt effects have *not* yet been fully included as a coupled 3D electrothermal field operator in V3.3. The nominal detector temperature is thus conditional on an important unresolved physical coupling. The complete 430-domain PDE has not been run and its global candidate mesh has substantial poor-quality regions.

This assessment retains actual CAD renderings, true ceramic fields, equations, transport profiles, full 88-leg/pad numerical boundary records, and explicit verification statistics. The older 38-page Rev3 is retained as a separate historical benchmark, rather than being confused with the current cooler.

Contents

Executive technical abstract **2**

1 Version identity and evidence hierarchy **4**

2 Replacement TEC physical architecture **4**

3 Shaded solid CAD inspection: the actual replacement geometry **6**

4 Series topology and copper parasitics **8**

5 Global geometric and meshing milestones **8**

6 Governing nonlinear thermoelectric model **9**

7 Original three-dimensional ceramic FEM **10**

8 Solder, lead and contact accounting **11**

9 Cavity and hot-side boundary assumptions **11**

10 Matched current reference cases **12**

11 Stage heat, voltage and temperature results **12**

12 Fixed-point convergence and independent accounting **13**

13 Engineering conclusion and next milestone **15**

14 Independent scientific reproduction **15**

15 Scientific and engineering references **15**

A Complete 14-knot P and N transport source tables **16**

B All original 88 P/N leg centers **17**

C Native ceramic-face geometric and solved heat ledger **19**

D Verification boundaries and remaining work **21**

1 Version identity and evidence hierarchy

Artifact	Subject	Verified status
Historical Rev3	Earlier 216-solid 28/12/4 cooler	Detailed field method; raw-gradient release failed
Replacement V3 R1	824 physical STEP bodies, formed tabs and joints	Conditional CAD design acceptance
R3.9 geometry	430 material regions, 2,440 CAD faces	Geometric mesh audit; conditioning unqualified
V3.3 simulation	88 nonlinear legs and four source 3D ceramic fields	One accepted numerical operating point; interstage metal physics incomplete

The cooler envelope is informed by TEC Microsystems MC04 family reference geometry; the package by a separate VIGO nine-pin TO-66 drawing. They are not identified as a commercially fabricated composite. The 29/11/4 stage split, material grades, solder properties and fabricated metal details are source-inspired or chosen synthetic hypotheses, *not* proprietary OEM internals.

2 Replacement TEC physical architecture

Four nominal ceramic plates have square footprints 6.4, 5.2, 3.8 and 2.4 mm, each 0.5 mm thick. Eighty-eight P/N legs are 0.4 by 0.4 by 0.5 mm each, grouped into 58, 22 and 8 legs from hot to cold. The package routing connects TEC supply/return pins 1/9 and separate HgCdTe photoconductor bias pins 7/8. The exported reference contains 824 CAD solids and 176 solder joints, 89 insulated TEC electrical nets, and formed internal metal links. The

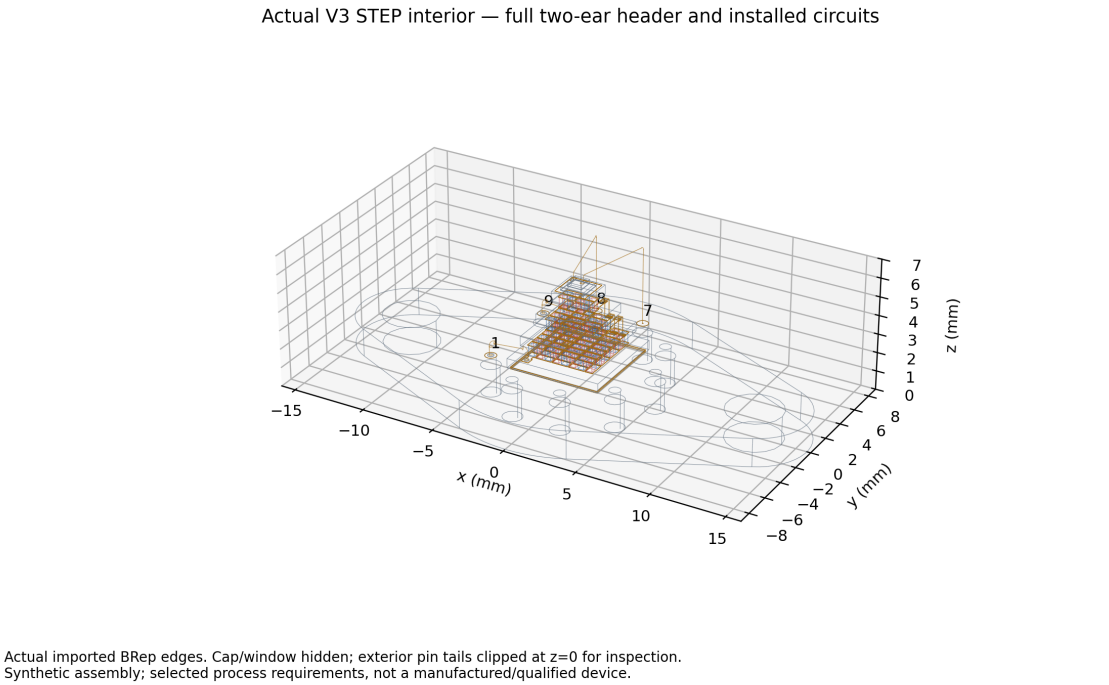


Figure 1: Genuine archived V3 replacement STEP package rendering. The package and cooler geometry are independently selected engineering references, not the geometry of one identified manufactured product.

original long, unsupported copper routing was rejected. The replacement has formed Cu-family foil tabs across nets 58 and 80, plus an extended three-foot return net 88 with a real middle thermal landing. Its design uses patterned ceramic metallization and finite solder footprints. The selected foil section is 0.300 by 0.030 mm; its

bend radius is 0.075 mm; finished foot joints are specified at 20 plus/minus 5 micrometres. No measured current rating or fatigue qualification follows.

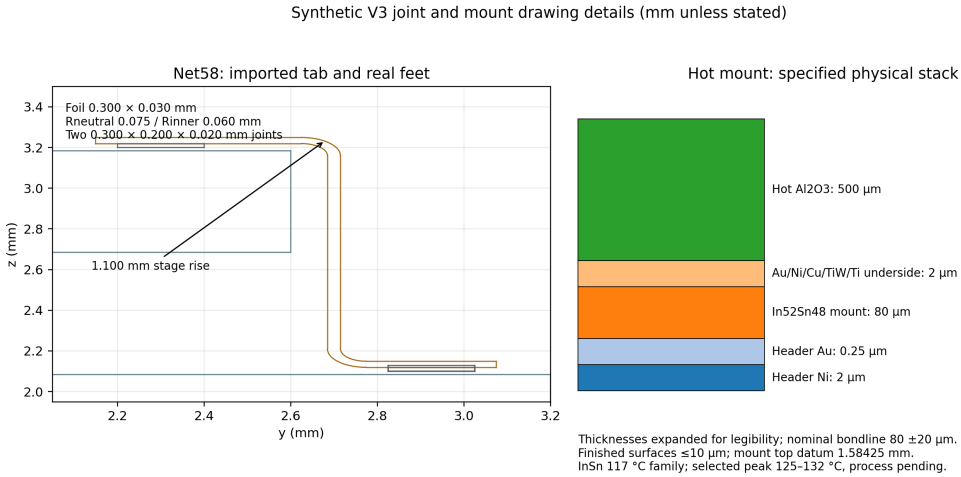


Figure 2: Actual CAD-derived joint, metallization and mounting sections. Joining tolerances and material/process choices remain selected synthetic requirements.

3 Shaded solid CAD inspection: the actual replacement geometry

The shaded views below tessellate the original 824-body source geometry using a true per-pixel depth buffer. The previous mean-face-depth renderer wrongly displayed hidden metallization through ceramic solids. This rendering is neither a reconstructed cooler nor an exploded assembly. All plates, P/N legs and analytically bent copper foil links retain the source positions, widths, heights, polarities and original path radii. Compared with the wireframe, the opaque solid bodies make the differences from the rejected long copper-post construction readily visible.

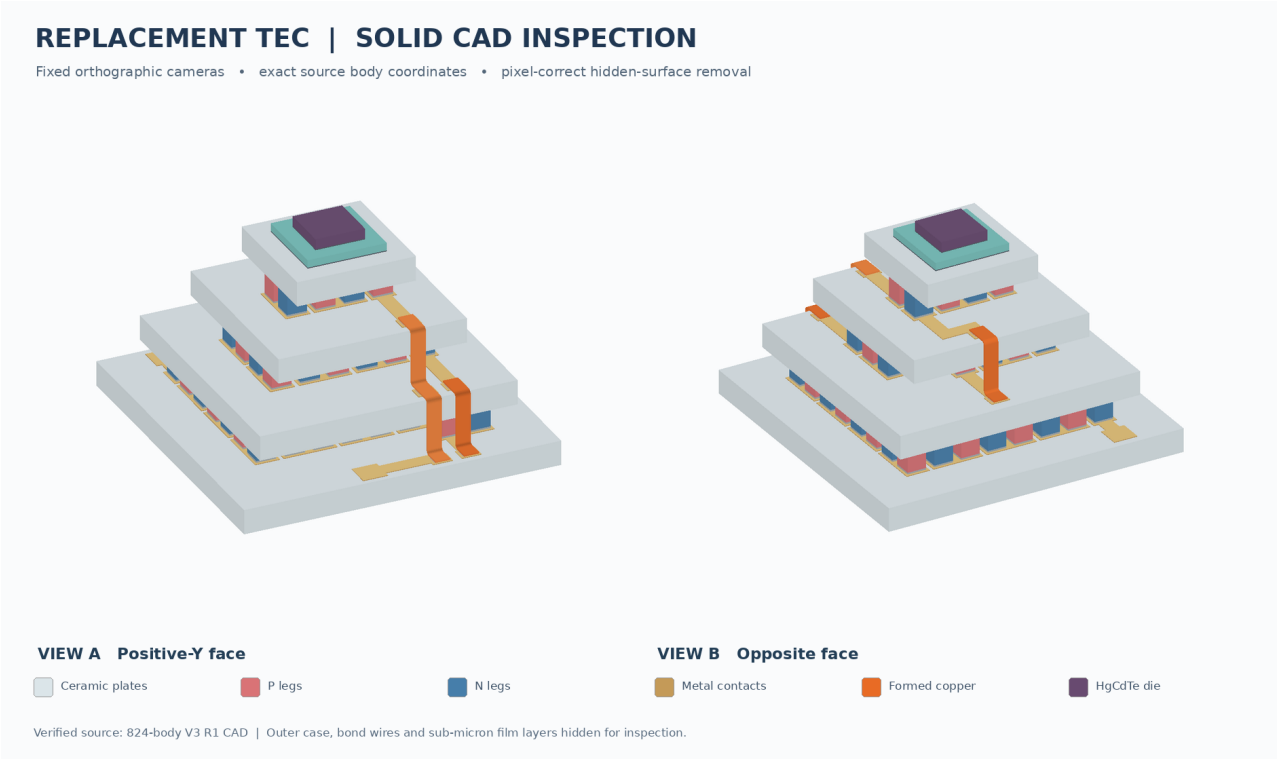


Figure 3: Correctly depth-occluded replacement geometry from two orthographic views: four ceramic stages, 58/22/8 P/N legs, visible exposed metallization only, and actual formed copper-foil connections. The external case, leads and very thin buried film layers are intentionally omitted. Original analytic foil-bend paths are sampled for shaded display, without relocating any CAD features.

3.1 Actual interstage foil cross-sections

The design uses two short supported formed copper-foil stage links (nets 58 and 80), and a longer stepped return (net 88). The last has two vertical steps and three physical feet, including an intermediate thermal landing. Unlike a generic connecting rod, its cross-section has nominal width 0.300 mm, thickness 0.030 mm and bend radius 0.075 mm. The foil is represented at actual thickness, not enlarged for visibility.

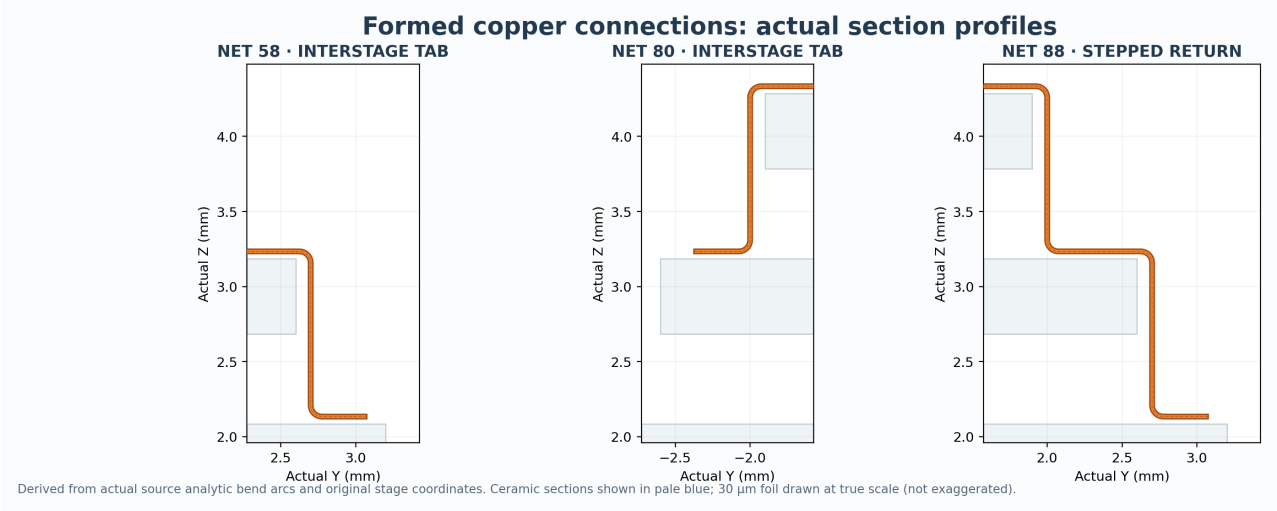


Figure 4: Y–Z sections reconstructed from the source analytic straight and circular-arc geometry at original coordinates, without exploded offsets. Pale blue regions indicate where ceramic cross-sections intersect each foil plane. The 30-micrometre copper thickness is drawn at true scale.

This inspection demonstrates the geometry change, but does not resolve missing formed-tab Joule heating, interface thermal resistance, electrothermal current crowding or interstage thermal bypass. Those remain necessary before the V3.3 detector-temperature estimate can qualify as a prediction for the assembled design.

4 Series topology and copper parasitics

The CAD conductor graph verifies all 88 legs as articulation vertices in the pin-1-to-pin-9 series circuit: removing any one leg disconnects the two terminals. Native transitions pass from P leg 57 to N leg 58 through tab 58, and from P79 to N80 through tab 80. The final leg P87 is connected to pin 9 through stepped return 88. These checks prove intended topology but not the temperature-dependent electrical or thermal properties of the connections. For a uniform slender conductor with temperature-dependent material properties,

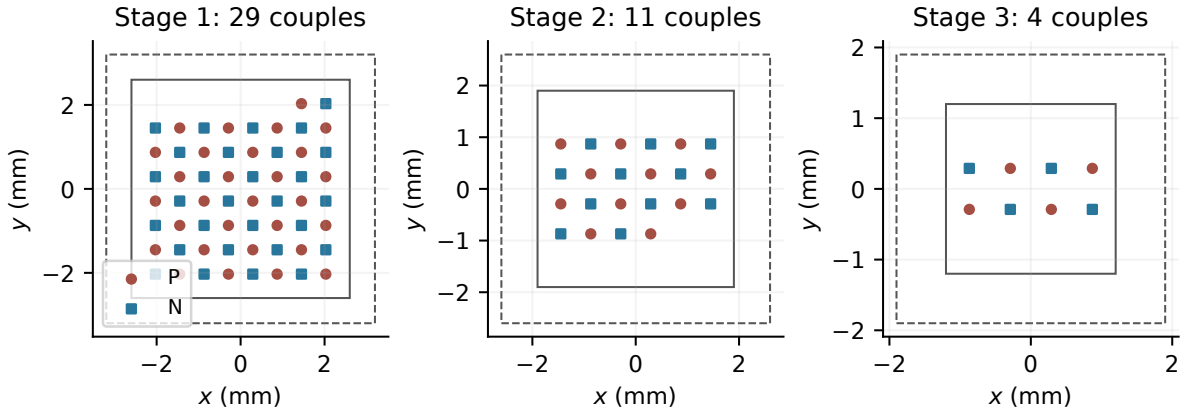


Figure 5: Actual saved stage-specific P/N leg centroids and plate footprints. This verifies the source positions used in the working hybrid; it does not depict a solved copper-tab thermal field.

$$R_e = \int_0^L \frac{\rho(T,s)}{A(s)}\,ds, \qquad G_{\mathrm{th}} = \left[\int_0^L \frac{ds}{k(T,s)A(s)}\right]^{-1}, \quad P_J = I^2 R_e.$$

The registered neutral path lengths are 1.96062 mm (net58), 1.97062 mm (net80), and 3.70624 mm (return88). Using a uniform 0.300 by 0.030 mm section, illustrative $\rho_{\mathrm{Cu}} = 1/(58\times10^6)\,\Omega\mathrm{m}$ and $k_{\mathrm{Cu}} = 385\,\mathrm{W/(m\,K)}$ yields the approximate geometry-only screen below. The competing design goals are explicit: reducing electrical resistance

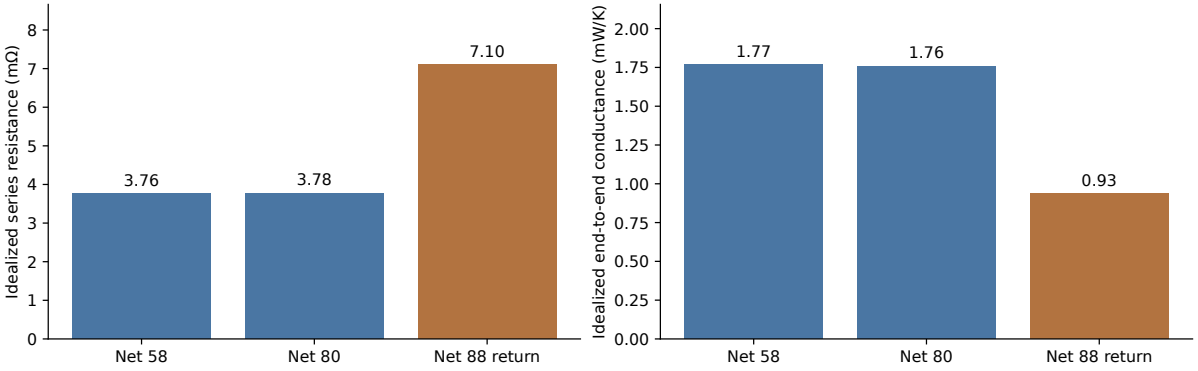


Figure 6: Derived idealized uniform-bar electrical and thermal copper coefficients. Actual formed-tab bends, patterned multilayer film, contact drops, variable temperatures, Joule partition and intermediate return landing are not resolved. The return is not correctly represented by a single two-terminal thermal conductance.

through a larger cross-section raises passive thermal coupling. A physical successor must solve or independently validate the full nonlocal stage-to-stage multiport heat and current network, including the return's intermediate land.

5 Global geometric and meshing milestones

The first full R3.8 TetGen volume candidate contained 1,043,314 tetrahedra over 430 material regions, with all 320,522 registered facets and 378 finite-contact witnesses retained. R3.9 applied 33,195 conservative 3-to-2 internal connectivity operations and reduced the mesh to 1,010,119 tetrahedra with unchanged original surface

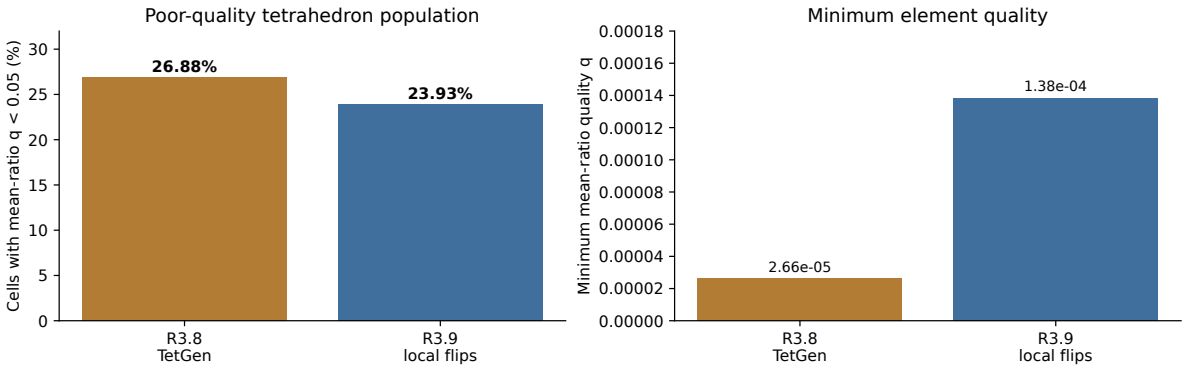


Figure 7: Independent archived global mesh-quality measurements. The 3-to-2 flips improved the poor-cell population, but the minimum quality and remaining weak cells prevent assuming mesh-converged coupled FEM.

facets and regional volumes. R3.9 retains 241,754 tetrahedra with mean-ratio quality below 0.05 (23.93 percent) and a minimum of 1.38495×10^{-4}. Thus no global coupled 430-domain electrical and thermal PDE is released. Historical full-solid electrical Gate B is FAIL/STOP; the first source N/P pair had eight raw gradient-current failures; the prospective P2 and global C/D computations are not qualified. Good energy closure in a separate hybrid model does not erase these original raw gates.

6 Governing nonlinear thermoelectric model

Each of the 88 axial P/N legs uses a signed 0.4 A series current and 32 longitudinal intervals. With z directed hot-to-cold, $S(T)$ Seebeck coefficient, $\rho(T)$ resistivity, $k(T)$ conductivity, and signed current density J, the constitutive conservation model is

$$q = S(T)TJ - k(T)\frac{dT}{dz}, \qquad \frac{dq}{dz} = J\left[\rho(T)J + S(T)\frac{dT}{dz}\right], \qquad \frac{dV}{dz} = \rho(T)J + S(T)\frac{dT}{dz}.$$

Midpoint constitutive evaluation and damped Newton iterations enforce thermal and electrical conservation at local temperatures, including the boundary identity $Q_h - Q_c = IV_{\text{leg}}$. The material source is a 14-knot figure digitization of research P-HE and N-HE-1 specimens over 40–300 K. Above 300 K the code uses an *unmeasured* one-quarter final-segment slope continuation; this is not a production material characterization.

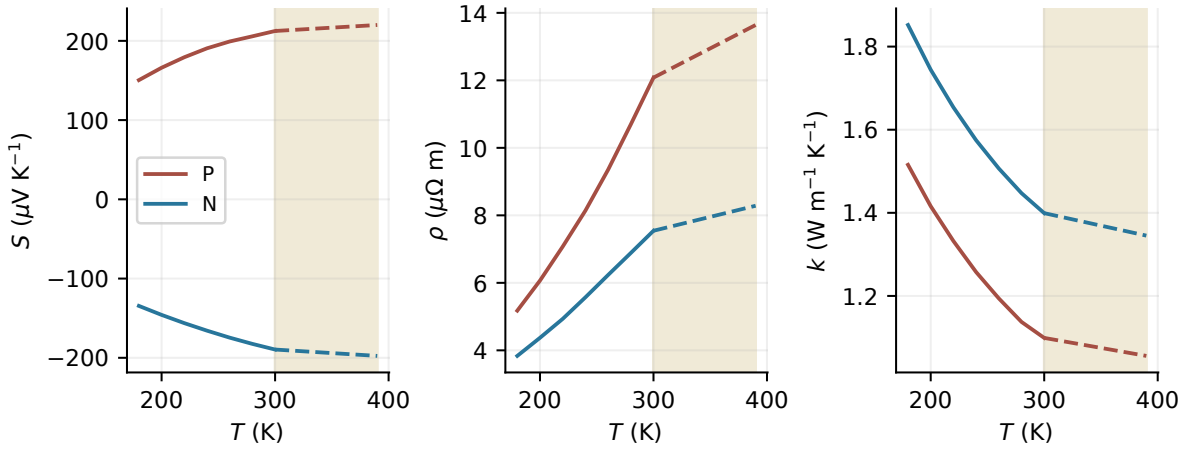


Figure 8: Source-derived P/N research analog properties. The shaded/dashed extension beyond 300 K is explicitly assumed; 58 hot-stage leg contacts exceed the research marker range.

7 Original three-dimensional ceramic FEM

Each ceramic independently solves

$$-\nabla \cdot (k_c \nabla T) = 0, \qquad K_{ab} = \sum_e \int_{\Omega_e} k_c \nabla N_a \cdot \nabla N_b \, d\Omega, \quad b_a = \sum_f \int_{A_f} N_a q_f \, dA,$$

with $k_c = 30\,\mathrm{W/(m\,K)}$ a selected constant-property alumina analogue. Original face-specific P/N leg heat and assumed joint/cavity contributions are distributed over native TRI3 pad faces; source-local pad means feed back into leg end temperatures. Four archived source meshes contain 4,936 / 4,085 / 1,986 / 784 TET4 elements, totaling 11,791 tetrahedra. The four fields share their temperature gauge and stage heat exchange with the

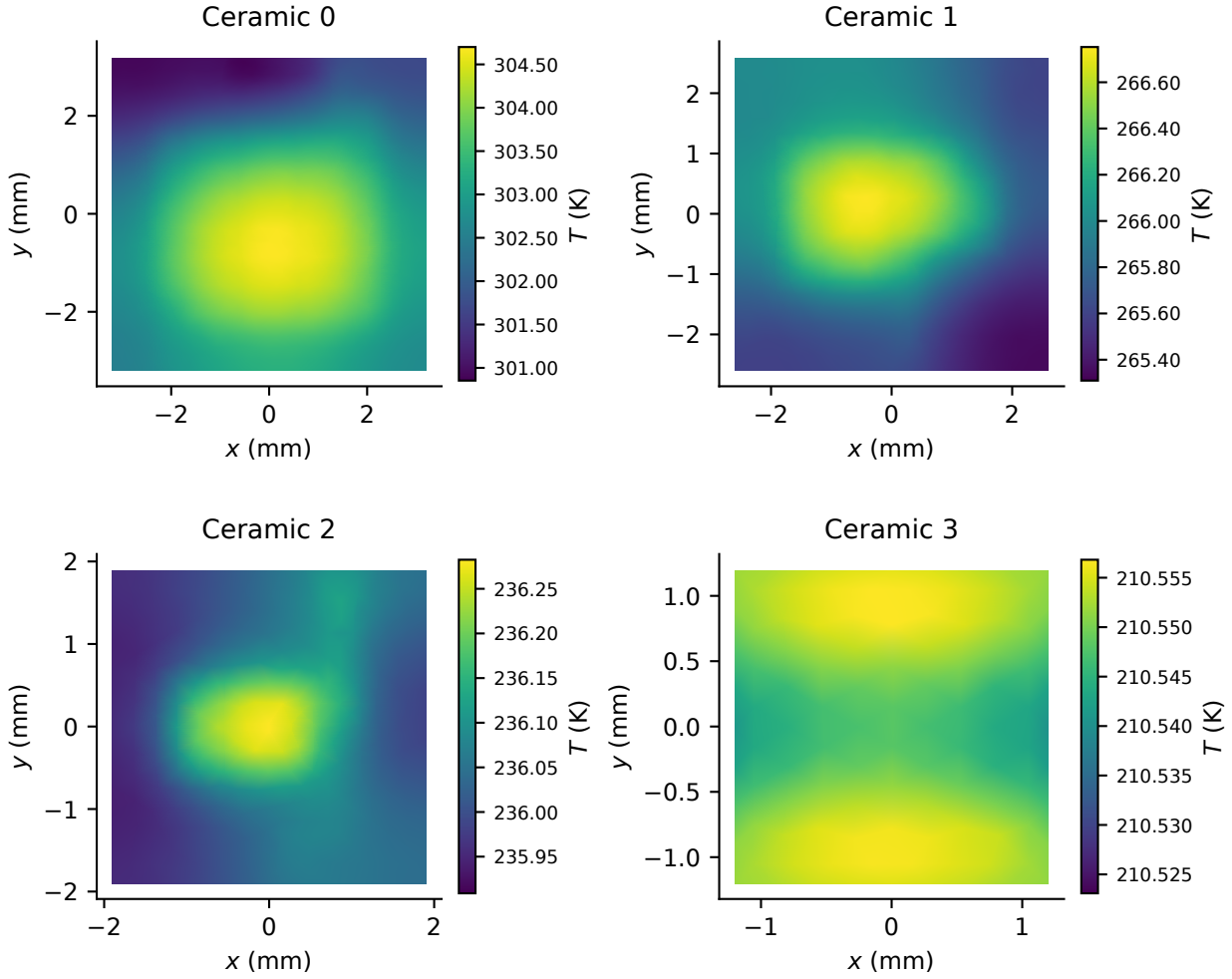


Figure 9: Real source-tetrahedron top-surface triangulations and solved nodal temperature fields from the original saved four ceramic archives. This is authentic spatial FEM evidence rather than a rendered approximation of the uncomputed full global package.

nonlocal reduced network. This is a single-mesh spatial result; no independent mesh-converged thermal accuracy bound is available.

8 Solder, lead and contact accounting

The current synthetic electrical model assumes 176 solder bodies of 35 micrometres thickness and 0.16 mm^2 area, with assumed SnSb resistivity $1.7 \times 10^{-7}\ \Omega\,\mathrm{m}$ and 352 assumed metal-interface specific resistances $r_c'' = 10^{-11}\ \Omega\,\mathrm{m}^2$. For each joint the resistance is

$$R_j = \rho_{\mathrm{SnSb}} \frac{L_j}{A_j} + 2\frac{r_c''}{A_j}, \quad R_{\mathrm{all}} = 0.028545\ \Omega, \quad I^2 R_{\mathrm{all}} = 4.56720\ \mathrm{mW}.$$

The implementation places 25.950 micowatts per source joint into its neighboring ceramic pad. This assumes the entire joint Joule heat reaches a ceramic and omits finite thermal-contact drops, interfacial thermopower and nonlocal copper-network heating. The four ceramic applied-joint terms, in mW, are [1.50510, 2.07600, 0.77850, 0.20760]. External supply Cu wires and detector Au bonds are separate effective electrical/thermal links. *They must not be mistaken for a resolved calculation of the newly formed tabs or return conductor.*

9 Cavity and hot-side boundary assumptions

The reference forward case imposes a 300 K cap/window/ambient thermal reservoir, a 100 Pa krypton environment, an effective 3.0 mm gas gap, accommodation 0.8, and nominal gas conductivity knots $k_{\mathrm{Kr}}(200, 300, 400\,\mathrm{K}) = (0.0064, 0.0095, 0.0122)\ \mathrm{W/(m\,K)}$. Gray radiation uses assumed emissivities $\epsilon_{\mathrm{cap}} = 0.20$, $\epsilon_{\mathrm{ceramic}} = 0.80$, and $\epsilon_{\mathrm{window}} = 0.10$, with selected effective view factors. The source effective exposed surface areas are [12.60, 8.68, 4.76, 1.00] mm^2. The generic exchange terms are

$$Q_{\mathrm{gas},i} = G_{\mathrm{eff,i}}(p,T)(T_\infty - T_i), \qquad Q_{\mathrm{rad},i} = \epsilon_{\mathrm{eff},i}\sigma A_i F_i (T_\infty^4 - T_i^4).$$

These are *effective engineering closures*, not cavity fluid dynamics, true free-molecular transport, exact exposed-face loads, spectral window modeling or full radiosity.

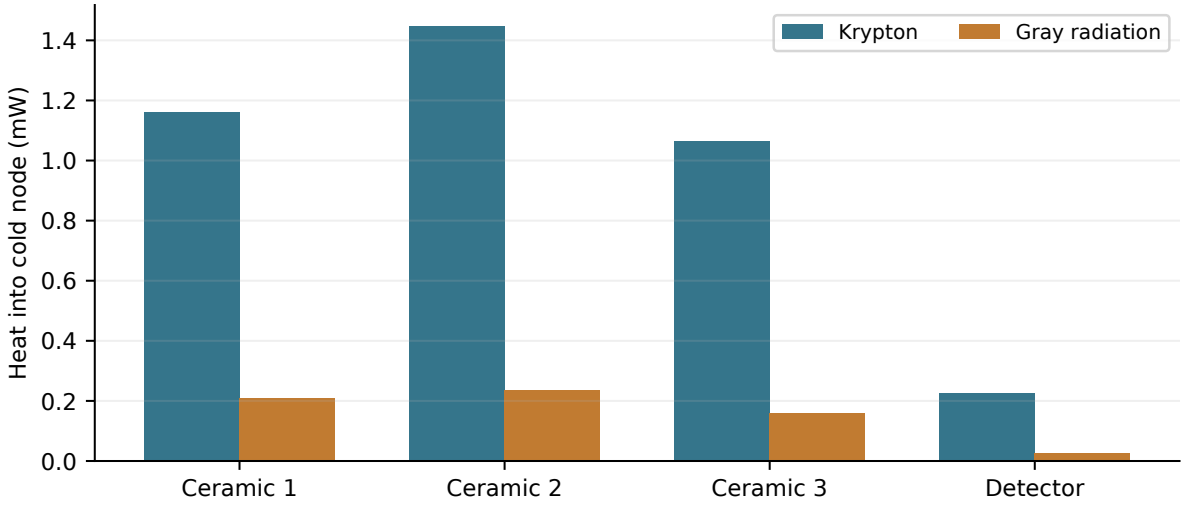


Figure 10: Archived V3.3 cavity contributions: total gas 3.892553 mW, gray radiation 0.624076 mW, combined heat into cold components 4.516629 mW. The reservoir carries an equal and opposite signed load.

Cold destination	Kr gas (mW)	Gray rad. (mW)	Total (mW)
Ceramic C1	1.15901	0.20874	1.36775
Ceramic C2	1.44533	0.23389	1.67923
Ceramic C3	1.06469	0.15770	1.22239
Detector	0.22352	0.02375	0.24726
TOTAL	3.89255	0.62408	4.51663

An effective header/TIM/radiator resistance of approximately 4.8334 K/W carries heat to the warm sink. Neither mount pressure nor material lot is an experimental datum. The cavity load is redistributed to native *upper metal-pad* sources in the current model; thermal gradients on exposed bare ceramic faces are not thereby demonstrated.

10 Matched current reference cases

The V3.2 baseline and V3.3 cavity-loaded run use the same 88-leg source geometry, 11,791 ceramic tetrahedra, 0.4 A input, 300 K reservoir, 2 mW detector load and assumed material/contact parameters. The cavity raises die temperature 0.637233 K, while the nonlinear terminal voltage changes slightly due to changed operating gradients.

Metric	V3.2 (cavity off)	V3.3 (cavity on)
Detector temperature (K)	209.987981	210.625214
Terminal voltage (V)	1.524071	1.522462
Electrical input (W)	0.609628	0.608985
Bath rejection (W)	0.590323	0.594044
Pressure (Pa)	0 (cavity off)	100 (Kr)

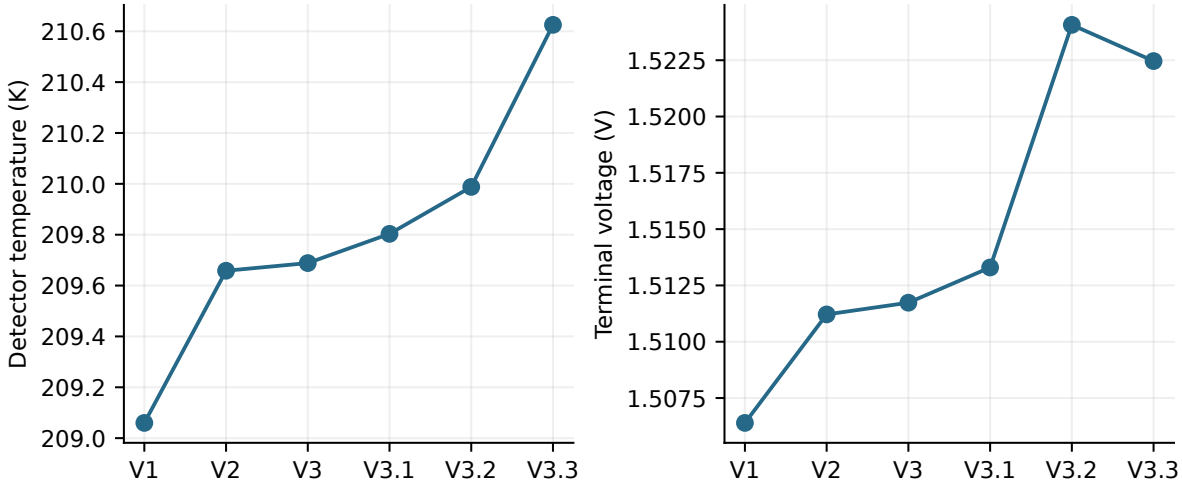


Figure 11: Recorded V1 to V3.3 complete reference-model milestones, all evaluated at the nominal current. These points describe changing model fidelity, not successive mesh convergence or independent measurements.

11 Stage heat, voltage and temperature results

Stage	Legs	Cold flux (mW)	Hot flux (mW)	V	Joule (mW)
1	58	179.9602	614.7396	1.086948	262.732
2	22	46.1636	176.5165	0.325882	81.390
3	8	5.1576	43.7058	0.096371	24.642

Coupled node	Temperature (K)
Stage0 Hot	303.65655
Stage0 Cold	265.90329
Stage1 Hot	266.34287
Stage1 Cold	236.00259
Stage2 Hot	236.15716
Stage2 Cold	210.52765
Carrier	210.55066
Die	210.62521

The external terminal voltage includes additional lead and joint contributions beyond summed P/N voltage. Stage energies must account for signed electrical work, source losses, detector heat, cavity sources and external sink. Directly adding cavity milliwatts to input electrical power without solving the changed stage temperatures would be incorrect.

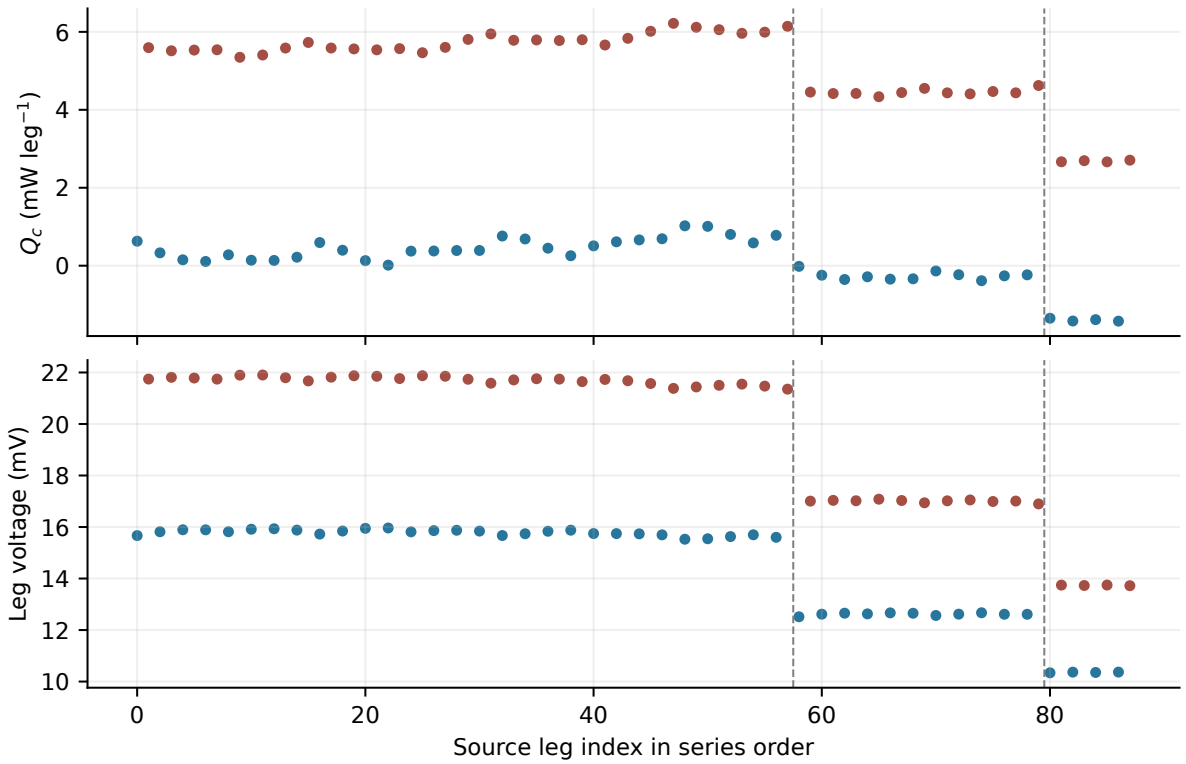


Figure 12: Real saved individual-leg cold-side heat and terminal-voltage contributions. Stage transitions occur after source leg indices 57 and 79. Source material laws and local ceramic pad temperatures determine leg-specific behavior.

12 Fixed-point convergence and independent accounting

The solver used the previous V3.2 solution as a warm start and iterated five spatial leg/pad updates, stopping at a maximum pad temperature change 1.65345×10^{-5} K below its registered 2×10^{-5} K criterion. That number is a *numerical stopping threshold*, not a temperature accuracy bound.

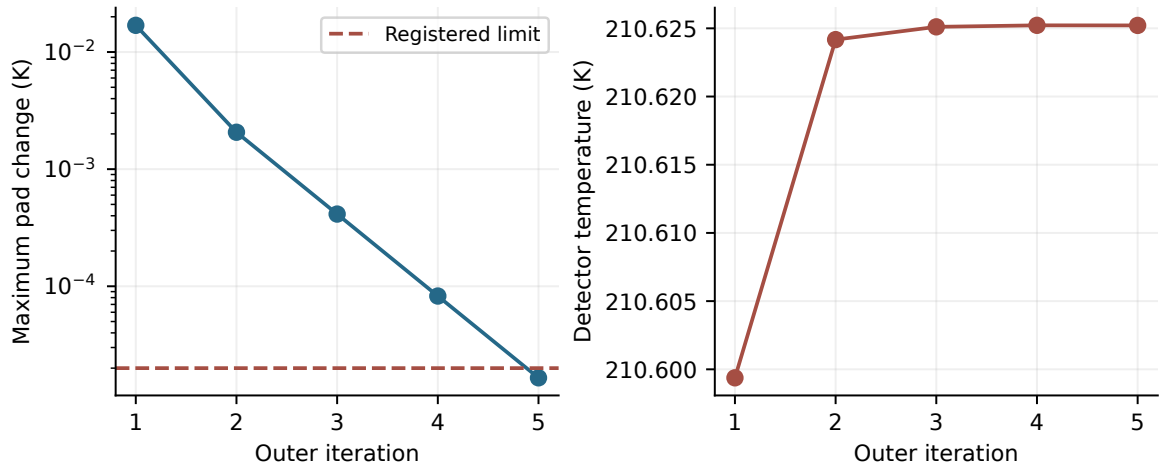


Figure 13: Actual saved spatial fixed-point iteration changes and detector temperature history. The last update meets the registered coarse convergence threshold; no mesh refinement or material uncertainty bound is inferred.

Iteration	Die T (K)	Max pad update (K)	Local flux mismatch (W)
1	210.599377	1.690e-02	3.914e-15
2	210.624173	2.066e-03	4.580e-16
3	210.625110	4.135e-04	1.478e-15
4	210.625220	8.254e-05	2.928e-15
5	210.625214	1.653e-05	5.503e-15

Metric	Absolute residual	Unit
Global energy	8.8674e-09	W
Worst stage balance	4.9061e-09	W
Ceramic pad heat	5.5025e-15	W
Worst individual leg	1.2364e-13	W
Final pad temperature change	1.6535e-05	K

An independent read-only field/load audit reconstructs the saved energy terms without rerunning the original PDE and finds a finite 2.535×10^{-8} W pad/leg update discrepancy. Numerical closure validates the *declared equations and discretization bookkeeping* at this operating point, not prediction fidelity of the as-built cooler.

13 Engineering conclusion and next milestone

The replacement cooler is a concrete 824-solid design with original STEP files, audited contact geometry and a stage-specific 430-region mesh. It is *not* necessary to restart cooler CAD development. The functioning V3.3 hybrid omits full nonlocal electrothermal field solutions for its formed tab58/tab80/return88 conductors, so the conditional temperature prediction cannot yet be called a full electrothermal result for the physical replacement. The next practical engineering milestone is a conservative electrical and thermal multiport extracted from this already constructed CAD, with named intermediate thermal landings and explicit film, solder and contact effects. The solver should then recompute the same 0.400 A / 300 K / 2 mW / 100 Pa nominal point and report changed die temperature, terminal voltage, stage heat budgets and sensitivity to uncertain contact inputs. Full 3D copper FEM may be justified if compact validated metal substructures cannot bound current crowding or nonlocal heat flow. The older 38-page Revision 3 is preserved because of its unusually useful FEM methodology and publication-level plots, *not* because it describes the current metal layout. The 43-page interim extension is historical, not the latest design reference.

14 Independent scientific reproduction

An independent implementation should use the four 0.5 mm-thick 6.4/5.2/3.8/2.4 mm ceramics, the Appendix B 88 leg centers and polarities, the 29/11/4-couple electrical series order and the source-selected 0.4 by 0.4 by 0.5 mm P/N geometry. Apply all six 14-knot P/N research-material curves in Appendix A; interpolate within the experimental range and explicitly impose the unmeasured quarter-final-segment-slope continuation above 300 K. Solve 88 individually signed 32-interval nonlinear axial thermoelectric legs and four source-geometry P1 ceramic steady conduction systems using local native TRI3 heat-face integrals. Set the declared assumed ceramic conductivity to 30 W/(m K), add the nominal 176 SnSb joints and 352 assumed electrical contacts, Cu and Au external leads, 2 mW detector heat, the stated finite mounting resistance and the effective 100 Pa Kr/gray radiation closure. Iterate local pad and leg flux/temperature feedback until the maximum change is below 0.000020 K and independently check all signed stage and total heat balances. The archived Appendix C face centroids, areas and flux totals permit a comparable heat-integral replication, not byte-exact nodal temperatures. Bitwise reproduction requires original tetrahedral connectivity, exact face polygon vertices, quadrature and solver conventions. No unpublished real-world detector measurements are claimed.

15 Scientific and engineering references

1. TEC Microsystems, three-stage 3MC04-044-05 cooler family datasheet (2019), external envelope and thermoelectric reference. Its hidden internal routing is unknown.

2. VIGO Photonics, 3TE TO-66 nine-pin housing reference, Rev. 9 (2024). The cooler and housing are independent reference products.

3. J. Zhang et al., low-temperature bismuth telluride thermoelectric measurement and error-analysis study, *Chinese Physics Letters* 42, 056304 (2025). DOI 10.1088/0256-307X/42/5/056304. Source P-HE and N-HE-1 specimens are not the actual installed module.

4. M. L. Huber, gas thermal conductivity tabulations (NIST/CRC, 2019): krypton material reference, not experimentally measured gap conduction.

5. TEC Microsystems and Kryotherm multistage construction drawings and photographs; Coherent mounting guidelines. These establish process families, not qualification of the selected synthetic formed-tab design.

A Complete 14-knot P and N transport source tables

The following 84 values comprise six figure-digitized research-specimen curves. They do not establish actual commercial material properties. The assumed extension beyond 300 K is

$$f(T) = f(300) + 0.25\,(T - 300)\frac{f(300) - f(280)}{20}, \qquad f \in \{S, \rho, k\}.$$

T K	S_P μV/K	ρ_P $\mu\Omega$ m	k_P W/(m K)	S_N μV/K	ρ_N $\mu\Omega$ m	k_N W/(m K)
40	31.477086	1.226120	2.776424	-39.762194	0.837035	3.928920
60	44.991486	1.550968	2.488935	-53.616397	1.156199	3.246955
80	60.679339	1.955036	2.260290	-67.469437	1.522950	2.816874
100	77.996604	2.446265	2.072406	-81.936140	1.920207	2.536709
120	96.944423	3.004830	1.909416	-95.992742	2.354830	2.310364
140	115.755946	3.567364	1.764537	-109.847998	2.820018	2.129254
160	133.684757	4.343818	1.630978	-122.547039	3.319204	1.973635
180	150.458980	5.175731	1.515526	-134.499967	3.835302	1.851971
200	166.078757	6.067058	1.415926	-145.840430	4.371811	1.744414
220	179.321421	7.073284	1.332165	-156.231255	4.932113	1.653884
240	190.594863	8.138932	1.257469	-165.737958	5.570441	1.574639
260	199.219885	9.351143	1.194083	-174.568277	6.232621	1.506729
280	205.739245	10.694094	1.137484	-182.445326	6.884545	1.447332
300	212.462550	12.080611	1.099006	-189.575244	7.543277	1.399227

B All original 88 P/N leg centers

Each leg has 0.1600 mm^2 area and 0.5000 mm axial length; stage index 0 is the hottest. All coordinates refer to the original selected synthetic CAD x/y frame in mm.

Series index	Stage	P/N	x (mm)	y (mm)
0	0	N	-2.030000	-2.030000
1	0	P	-1.450000	-2.030000
2	0	N	-0.870000	-2.030000
3	0	P	-0.290000	-2.030000
4	0	N	0.290000	-2.030000
5	0	P	0.870000	-2.030000
6	0	N	1.450000	-2.030000
7	0	P	2.030000	-2.030000
8	0	N	2.030000	-1.450000
9	0	P	1.450000	-1.450000
10	0	N	0.870000	-1.450000
11	0	P	0.290000	-1.450000
12	0	N	-0.290000	-1.450000
13	0	P	-0.870000	-1.450000
14	0	N	-1.450000	-1.450000
15	0	P	-2.030000	-1.450000
16	0	N	-2.030000	-0.870000
17	0	P	-1.450000	-0.870000
18	0	N	-0.870000	-0.870000
19	0	P	-0.290000	-0.870000
20	0	N	0.290000	-0.870000
21	0	P	0.870000	-0.870000
22	0	N	1.450000	-0.870000
23	0	P	2.030000	-0.870000
24	0	N	2.030000	-0.290000
25	0	P	1.450000	-0.290000
26	0	N	0.870000	-0.290000
27	0	P	0.290000	-0.290000
28	0	N	-0.290000	-0.290000
29	0	P	-0.870000	-0.290000
30	0	N	-1.450000	-0.290000
31	0	P	-2.030000	-0.290000
32	0	N	-2.030000	0.290000
33	0	P	-1.450000	0.290000
34	0	N	-0.870000	0.290000
35	0	P	-0.290000	0.290000
36	0	N	0.290000	0.290000
37	0	P	0.870000	0.290000
38	0	N	1.450000	0.290000
39	0	P	2.030000	0.290000
40	0	N	2.030000	0.870000
41	0	P	1.450000	0.870000
42	0	N	0.870000	0.870000
43	0	P	0.290000	0.870000
44	0	N	-0.290000	0.870000
45	0	P	-0.870000	0.870000
46	0	N	-1.450000	0.870000
47	0	P	-2.030000	0.870000
48	0	N	-2.030000	1.450000
49	0	P	-1.450000	1.450000
50	0	N	-0.870000	1.450000
51	0	P	-0.290000	1.450000
52	0	N	0.290000	1.450000
53	0	P	0.870000	1.450000
54	0	N	1.450000	1.450000
55	0	P	2.030000	1.450000
56	0	N	2.030000	2.030000
57	0	P	1.450000	2.030000
58	1	N	1.450000	0.870000
59	1	P	0.870000	0.870000
60	1	N	0.290000	0.870000
61	1	P	-0.290000	0.870000
62	1	N	-0.870000	0.870000
63	1	P	-1.450000	0.870000
64	1	N	-1.450000	0.290000
65	1	P	-0.870000	0.290000
66	1	N	-0.290000	0.290000
67	1	P	0.290000	0.290000
68	1	N	0.870000	0.290000
69	1	P	1.450000	0.290000
70	1	N	1.450000	-0.290000
71	1	P	0.870000	-0.290000
72	1	N	0.290000	-0.290000
73	1	P	-0.290000	-0.290000
74	1	N	-0.870000	-0.290000
75	1	P	-1.450000	-0.290000
76	1	N	-1.450000	-0.870000
77	1	P	-0.870000	-0.870000

Series index	Stage	P/N	x (mm)	y (mm)
78	1	N	-0.290000	-0.870000
79	1	P	0.290000	-0.870000
80	2	N	0.870000	-0.290000
81	2	P	0.290000	-0.290000
82	2	N	-0.290000	-0.290000
83	2	P	-0.870000	-0.290000
84	2	N	-0.870000	0.290000
85	2	P	-0.290000	0.290000
86	2	N	0.290000	0.290000
87	2	P	0.870000	0.290000

C Native ceramic-face geometric and solved heat ledger

For each original ceramic and native thermal face, the table lists source face ID, centroid (mm), original CAD area (mm^2) and the signed integrated solved heat *into* that ceramic (W). Contributions include local P/N transport, provisional joint heating, finite sink, detector and effective cavity loads when applicable. Exact face polygons are not fully specified by centroids and areas.

Cer.	Face ID	x	y	z	Area mm^2	Heat in (W)
0	6	0.000000	0.000000	1.584250	37.21000000	-5.9404375264e-01
0	8	-2.030000	-2.415000	2.084250	0.58100000	+6.9212449542e-03
0	9	-1.160000	-2.030000	2.084250	0.54000000	+2.1001118011e-02
0	10	-0.000000	-2.030000	2.084250	0.54000000	+2.0799385661e-02
0	11	-2.030000	-1.160000	2.084250	0.54000000	+2.1333430585e-02
0	12	-1.160000	-1.450000	2.084250	0.54000000	+2.0925106301e-02
0	13	-1.160000	-0.870000	2.084250	0.54000000	+2.1098181465e-02
0	14	-2.030000	-0.000000	2.084250	0.54000000	+2.1661444954e-02
0	15	-1.160000	-0.290000	2.084250	0.54000000	+2.1280527762e-02
0	16	-0.000000	-1.450000	2.084250	0.54000000	+2.0726535211e-02
0	17	0.000000	-0.870000	2.084250	0.54000000	+2.0872339885e-02
0	18	-0.000000	-0.290000	2.084250	0.54000000	+2.1135046258e-02
0	19	1.160000	-2.030000	2.084250	0.54000000	+2.0762274895e-02
0	20	2.030000	-1.740000	2.084250	0.54000000	+2.0894962316e-02
0	21	1.160000	-1.450000	2.084250	0.54000000	+2.0664089207e-02
0	22	1.160000	-0.870000	2.084250	0.54000000	+2.0730610371e-02
0	23	1.160000	-0.290000	2.084250	0.54000000	+2.0987932670e-02
0	24	2.030000	-0.580000	2.084250	0.54000000	+2.1025501033e-02
0	25	-1.160000	0.290000	2.084250	0.54000000	+2.1501510390e-02
0	26	-2.030000	1.160000	2.084250	0.54000000	+2.2056807823e-02
0	27	-1.160000	0.870000	2.084250	0.54000000	+2.1665224003e-02
0	28	-1.160000	1.450000	2.084250	0.54000000	+2.1972015528e-02
0	29	0.000000	0.290000	2.084250	0.54000000	+2.1329922044e-02
0	30	-0.000000	0.870000	2.084250	0.54000000	+2.1513476612e-02
0	31	0.000000	1.450000	2.084250	0.54000000	+2.1762448001e-02
0	32	1.160000	0.290000	2.084250	0.54000000	+2.1132384800e-02
0	33	2.030000	0.580000	2.084250	0.54000000	+2.1315494607e-02
0	34	1.160000	0.870000	2.084250	0.54000000	+2.1316483722e-02
0	35	1.160000	1.450000	2.084250	0.54000000	+2.1497446937e-02
0	36	0.088570	2.858995	2.084250	0.67225000	-2.2200926469e-02
0	37	2.030000	1.740000	2.084250	0.54000000	+2.1652376653e-02
0	38	1.450000	2.379622	2.084250	0.51850000	+1.4709356454e-02
1	44	-1.740000	-2.030000	2.684250	0.54000000	-6.1717204457e-03
1	45	-0.580000	-2.030000	2.684250	0.54000000	-5.7923394592e-03
1	46	0.580000	-2.030000	2.684250	0.54000000	-5.6313481447e-03
1	47	-1.740000	-1.450000	2.684250	0.54000000	-5.8945158835e-03
1	48	-1.740000	-0.870000	2.684250	0.54000000	-6.1272498337e-03
1	49	-0.580000	-1.450000	2.684250	0.54000000	-5.6677985879e-03
1	50	-0.580000	-0.870000	2.684250	0.54000000	-5.9073535857e-03
1	51	-1.740000	-0.290000	2.684250	0.54000000	-6.2861601984e-03
1	52	-1.740000	0.290000	2.684250	0.54000000	-6.4929448577e-03
1	53	-0.580000	-0.290000	2.684250	0.54000000	-6.1441766581e-03
1	54	0.580000	-1.450000	2.684250	0.54000000	-5.4949366062e-03
1	55	0.580000	-0.870000	2.684250	0.54000000	-5.6165637076e-03
1	56	0.580000	-0.290000	2.684250	0.54000000	-5.9284010470e-03
1	57	1.740000	-2.030000	2.684250	0.54000000	-5.5964450493e-03
1	58	1.740000	-1.450000	2.684250	0.54000000	-5.5750930796e-03
1	59	1.740000	-0.870000	2.684250	0.54000000	-5.5316729855e-03
1	60	1.740000	-0.290000	2.684250	0.54000000	-5.7862099653e-03
1	61	-0.580000	0.290000	2.684250	0.54000000	-6.4257386885e-03
1	62	-1.740000	0.870000	2.684250	0.54000000	-6.8576261766e-03
1	63	-1.740000	1.450000	2.684250	0.54000000	-7.0890067759e-03
1	64	-0.580000	0.870000	2.684250	0.54000000	-6.6249732312e-03
1	65	-0.580000	1.450000	2.684250	0.54000000	-7.0137372522e-03
1	66	0.580000	0.290000	2.684250	0.54000000	-6.1746390641e-03
1	67	0.580000	0.870000	2.684250	0.54000000	-6.3948651910e-03
1	68	0.580000	1.450000	2.684250	0.54000000	-6.7126784447e-03
1	69	1.740000	0.290000	2.684250	0.54000000	-6.0016216656e-03
1	70	1.740000	0.870000	2.684250	0.54000000	-6.1198269805e-03
1	71	1.740000	1.450000	2.684250	0.54000000	-6.5247521871e-03
1	72	1.740000	2.030000	2.684250	0.54000000	-6.8707555753e-03
1	74	0.870000	2.300000	3.184250	0.12000000	+2.3941388903e-05
1	75	1.450000	1.470862	3.184250	0.67900000	+5.1504788556e-03
1	76	-1.450000	-0.580000	3.184250	0.54000000	+1.6210670281e-02
1	77	-0.580000	-0.870000	3.184250	0.54000000	+1.6207954022e-02
1	78	-0.580000	-0.290000	3.184250	0.54000000	+1.6071664242e-02
1	79	0.290000	-1.435162	3.184250	0.65650000	+1.1539642240e-02
1	80	0.580000	-0.290000	3.184250	0.54000000	+1.6217492734e-02
1	81	1.450000	0.000000	3.184250	0.54000000	+1.6376311378e-02
1	82	-1.450000	0.580000	3.184250	0.54000000	+1.6153119097e-02
1	83	-0.580000	0.290000	3.184250	0.54000000	+1.6048683270e-02
1	84	-0.580000	0.870000	3.184250	0.54000000	+1.6098735692e-02
1	85	0.580000	0.290000	3.184250	0.54000000	+1.6138044094e-02
1	86	0.580000	0.870000	3.184250	0.54000000	+1.6218414034e-02

Cer.	Face ID	x	y	z	Area	Heat (W)
2	92	-1.160000	-0.870000	3.784250	0.54000000	-4.1229763184e-03
2	93	-1.160000	-0.290000	3.784250	0.54000000	-4.0325885849e-03
2	94	0.000000	-0.870000	3.784250	0.54000000	-4.3372223253e-03
2	95	-0.000000	-0.290000	3.784250	0.54000000	-4.1255544021e-03
2	96	1.160000	-0.290000	3.784250	0.54000000	-4.2477066490e-03
2	97	-1.160000	0.290000	3.784250	0.54000000	-3.9991911145e-03
2	98	-1.160000	0.870000	3.784250	0.54000000	-4.0132490723e-03
2	99	0.000000	0.290000	3.784250	0.54000000	-4.0452781812e-03
2	100	1.160000	0.290000	3.784250	0.54000000	-4.1617896577e-03
2	101	-0.000000	0.870000	3.784250	0.54000000	-4.1215621922e-03
2	102	1.160000	0.870000	3.784250	0.54000000	-4.3855401752e-03
2	104	0.870000	0.829062	4.284250	0.64000000	+8.5720985048e-03
2	105	0.692720	-0.929413	4.284250	0.81350000	+3.2580848072e-03
2	106	-0.000000	-0.290000	4.284250	0.54000000	+1.1238504436e-02
2	107	-0.870000	0.000000	4.284250	0.54000000	+1.1290111066e-02
2	108	0.000000	0.290000	4.284250	0.54000000	+1.1233859859e-02
3	114	0.580000	0.290000	4.884250	0.54000000	-1.2327805666e-03
3	115	-0.580000	-0.290000	4.884250	0.54000000	-1.2229792254e-03
3	116	0.580000	-0.290000	4.884250	0.54000000	-1.2662612022e-03
3	117	-0.580000	0.290000	4.884250	0.54000000	-1.2279447029e-03
3	119	-0.000000	-0.000000	5.384250	4.41000000	+4.9499656971e-03

D Verification boundaries and remaining work

Technical claim	Actual status	Appropriate requirement for improvement
824-solid replacement V3 CAD	Conditional physical-design PASS	Measured forming, solder, current and reliability capability
Native pin-to-pin topology	88/88 legs PASS	Finite real current density, resistance and contact temperature fields
Global 430-domain R3.9 mesh	Geometric PASS	Improve worst elements and prove solver conditioning/convergence
Original full-solid electrical Gate B	Historical FAIL/STOP	Qualify original raw-gradient tests on a viable mesh
Full 430-domain electrothermal PDE	NOT RUN	Integrated metal/contact/leg/ceramic field solution
V3.3 hybrid at 0.4 A	Numerical PASS	Copper multiport closure, material sensitivity and current/load response
Rarefied gas and radiation	Effective load model	Native exposed-surface gas/gray/spectral physics
Real thermoelectric grades	Unidentified	Bounded synthetic studies or actual module certificates
HgCdTe photodetection	No carrier/optical PDE	Detector noise/spectral/bias physics if needed
As-built product prediction	Unvalidated	Device measurements; do not claim manufacturer equivalence

Publication policy. The original 38-page Revision 3 remains a sealed separate historical methodology reference. Its later 43-page interim source-mesh integration is likewise archived. This document describes the newer replacement CAD and the most recent completed running hybrid with the unmodeled internal copper paths prominently disclosed. All figures and numerical tables derive from existing saved evidence, and no new physics calculation was performed for this publication.